Friedrich-Alexander-Universität
Technische Fakultät

Welcome

CNN-Based Estimation of Sagittal Plane Walking and Running Biomechanics from Measured and Simulated Inertial Sensor Data

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Agenda

01 Introduction
- Problem Overview
- Main Idea

02 Methods
- The Simulation
- CNN Model Overview

03 Results

04 Discussion

05 Conclusion

06 Proposed Approach

07 References

Machine Learning and Data Analytics Lab (MaD) in Erlangen

Introduction

Authors:

- Anne D. Koelewijn
- Chris F. Martindale
- Eva Dorschky
- Anton J. Bogert
- Miriam Nitschke
- Bjoern M. Eskofier

- Deep learning needs a lot of data
- The authors generate extra IMU data using a biomechanical simulation
- This creates virtual training examples so the model can learn better

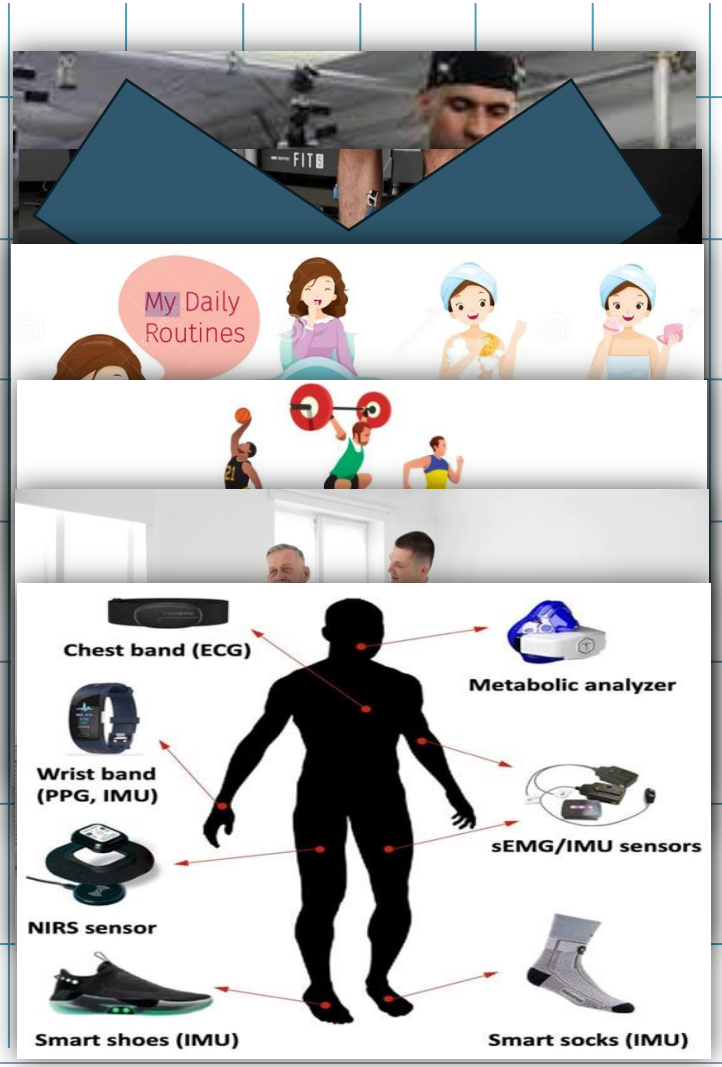
Problem Overview

- The Problem is – Lack of Data
- Biomechanics normally requires:
  - Optical Motion Capture (OMC)
  - Force plates
- These systems are:
  - Very expensive
  - Not portable
  - Only usable in a laboratory

✓ Everyday life ✓ Sports ✓ Rehabilitation

○ Wearable sensors (IMUs)

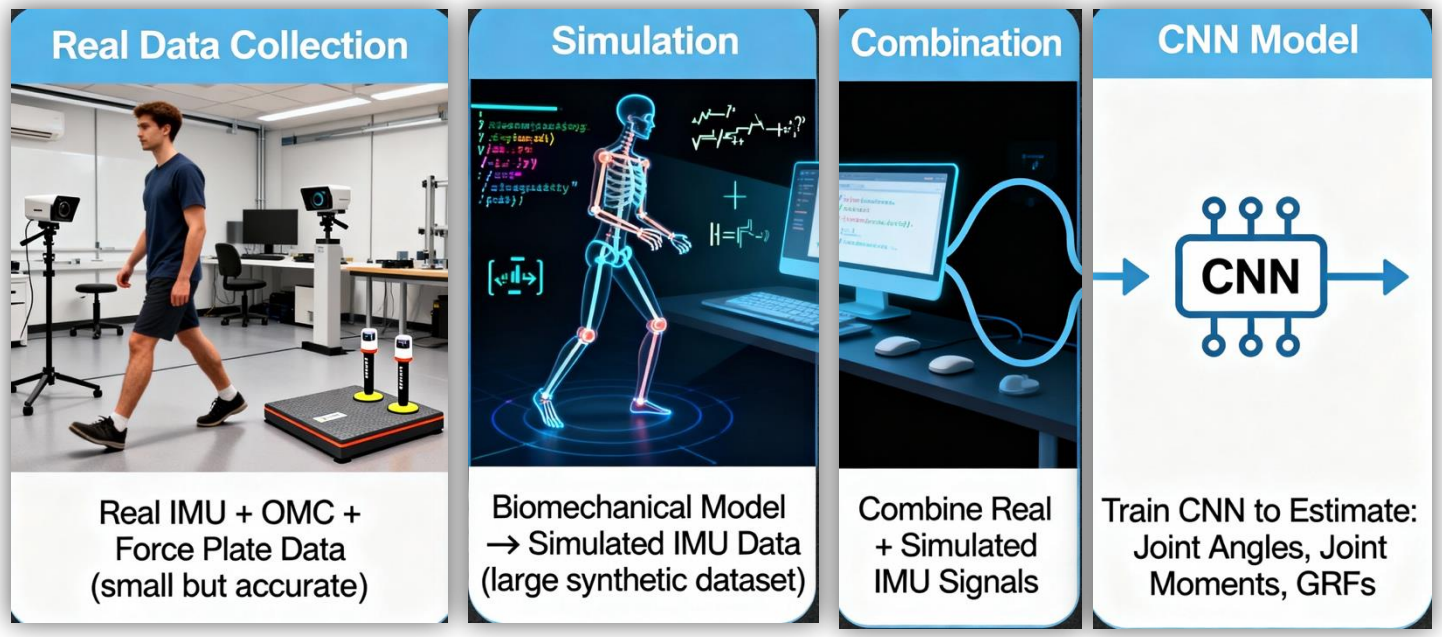
BUT: IMU datasets are **too small** for deep learning models



Main Idea

➢ How do they generate simulated IMU data?

- Collect Real Measurements:
  - Use OMC, IMUs, and force plates
- Generate Simulated Data
- Combine Both:
  - Real IMU data (small, high-quality)
  - Simulated IMU data (large, synthetic)



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- Train a CNN Using Both Inputs to Estimate:
  - Joint angles
  - Joint moments
  - Ground reaction forces (GRFs)

The Simulation

➢ Generating Simulated IMU Data

- ✓ Start with Real Measurements
  - 10 subjects walking & running at different speeds
- ✓ Random Sampling of Biomechanical Variables
- ✓ For each subject, these values are generated using a **multivariate Gaussian distribution**
- ✓ Create many new, realistic movement patterns

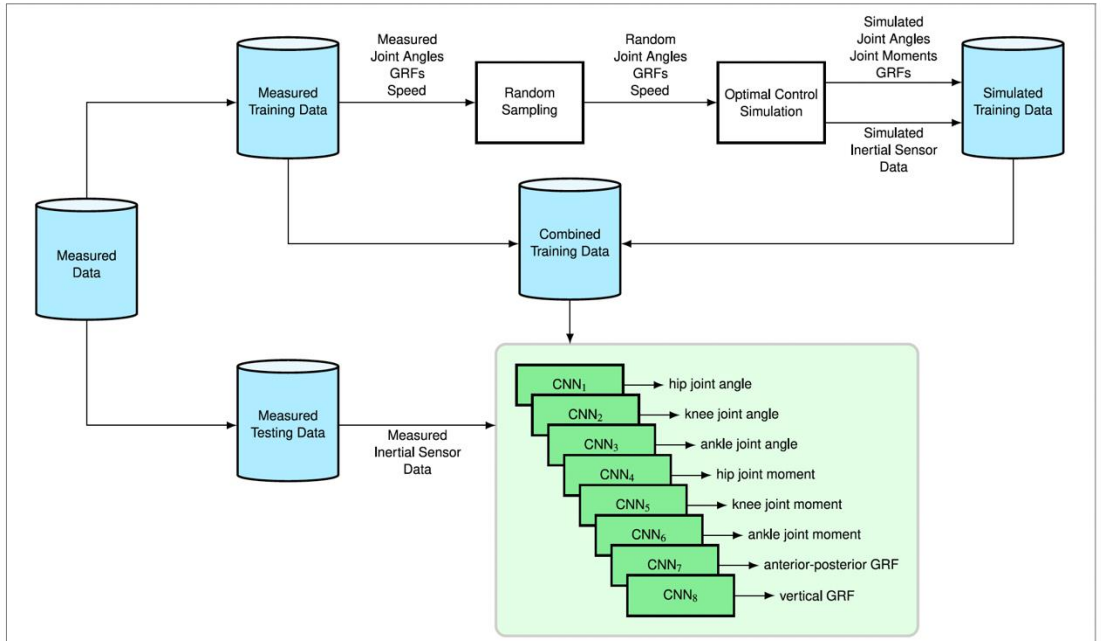
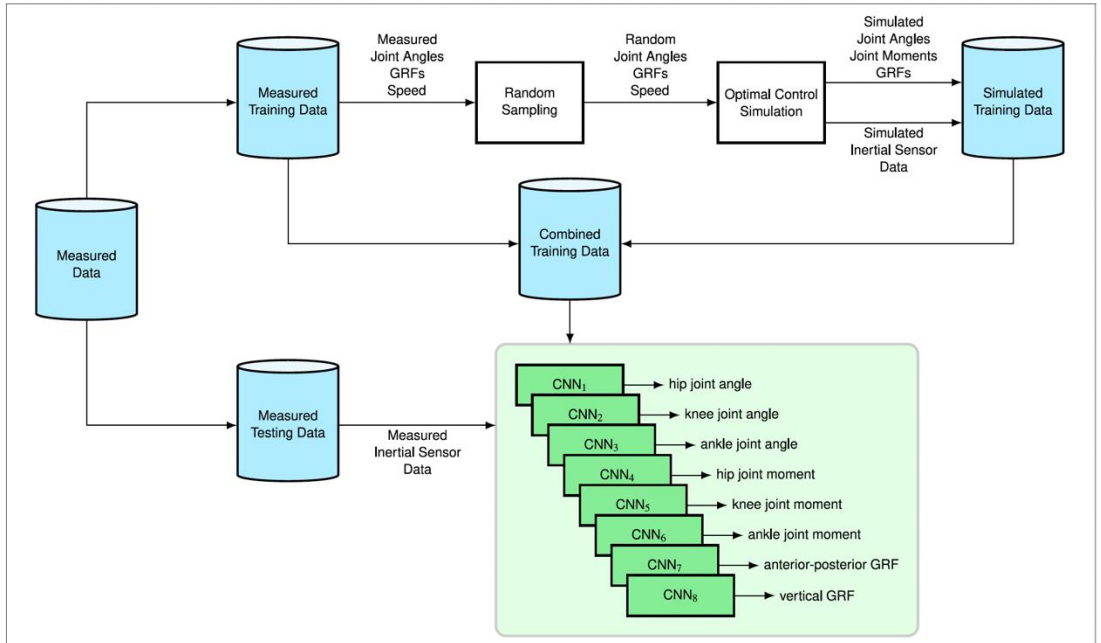


FIGURE 1 | We trained CNNs to estimate sagittal lower body kinematics and kinetics from accelerometer and gyroscope data from four inertial sensors which were placed on the lower body. Therefore, we created simulated data based on the measured training dataset (described in section 2.1): we drew random samples from measured joint angles, GRFs, and walking/running speeds (see section 2.2), which were then tracked by musculoskeletal models solving optimal control problems (see section 2.3). Simulated movements yielded biomechanics with matching inertial sensor data using a virtual inertial sensor model.

Dorschky et al., Frontiers in Bioengineering and Biotechnology, 2020

The Simulation

- ✓ Full-Body Musculoskeletal Simulation
  - Use a 2D musculoskeletal model
- ✓ Generation of Virtual IMU Signals
  - Place virtual sensors in the same locations as the real IMUs
  - Produces large realistic synthetic IMU datasets



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Dorschky et al., Frontiers in Bioengineering and Biotechnology, 2020

- ✓ Outcome: A big set of real + simulated IMU data to train the CNN model

CNN Model Overview

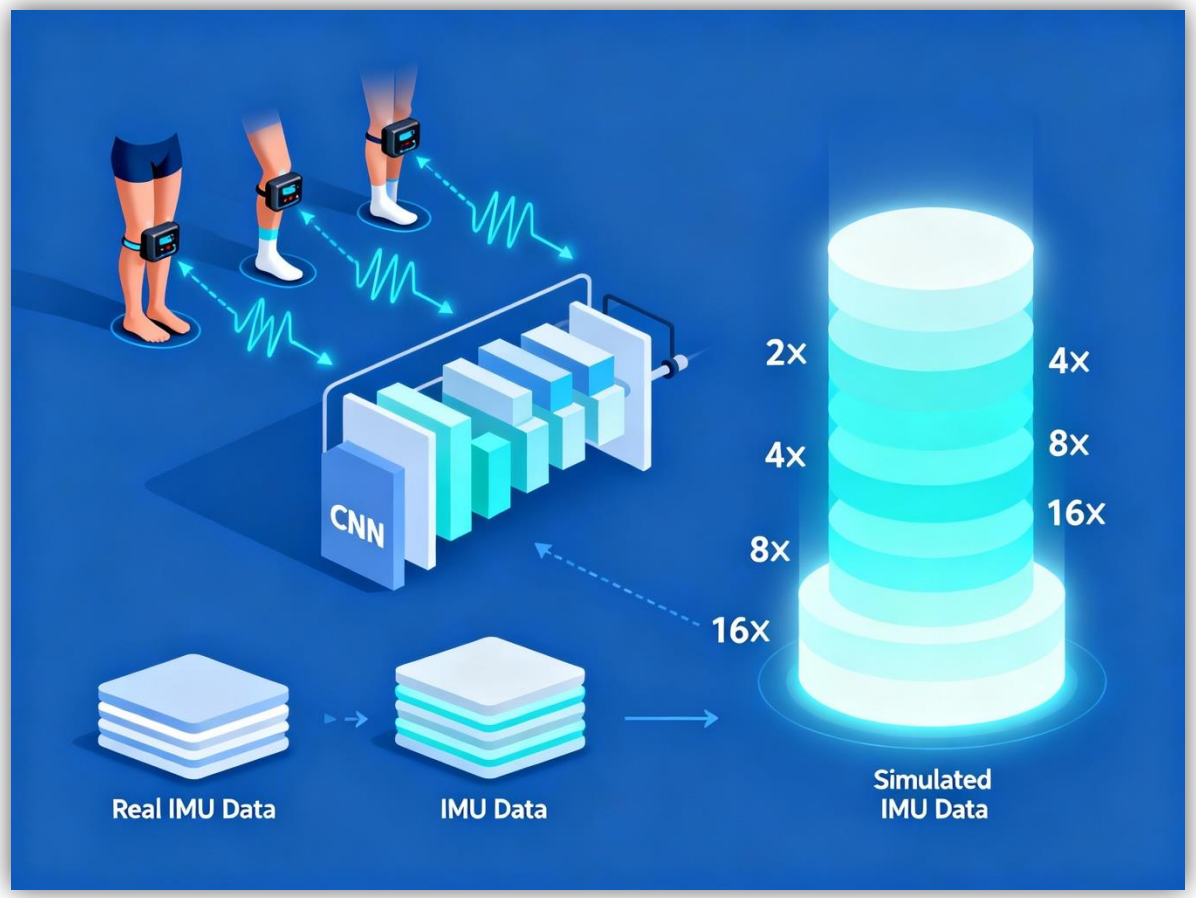
✓ CNN to estimate key biomechanical variables from IMU data

✓ Model Inputs

- IMU signals from 4 wearable sensors (6 channels per sensor)
- Add noise, to make the fake IMU look more realistic

Idea for future work

- Testing different combinations:
  - Real only
  - Real + 2× simulated
  - Real + 4× simulated
  - Real + 8× simulated
  - Real + 16× simulated



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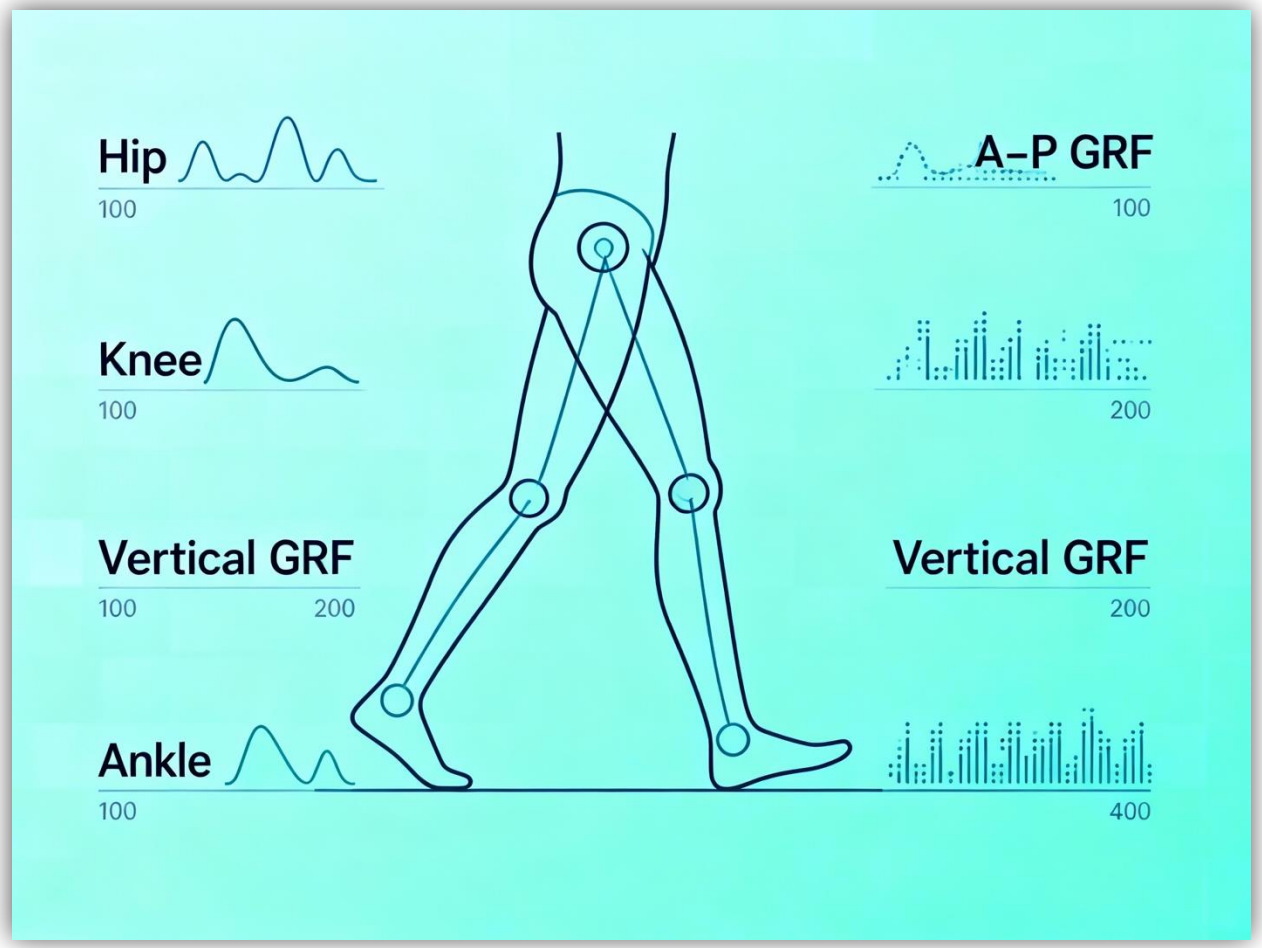
✓ How adding simulated data affects the model performance

CNN Model Overview

✓ Model Outputs

- Predicts 8 biomechanical variables (each over 100 time points *of one gait cycle*)
  - Joint Angles
    - Hip
    - Knee
    - Ankle
  - Joint Moments
  - Ground Reaction Forces
    - Anterior–Posterior GRF
    - Vertical GRF
- So, from IMU data only, the model can estimate both kinematics and kinetics during walking and running



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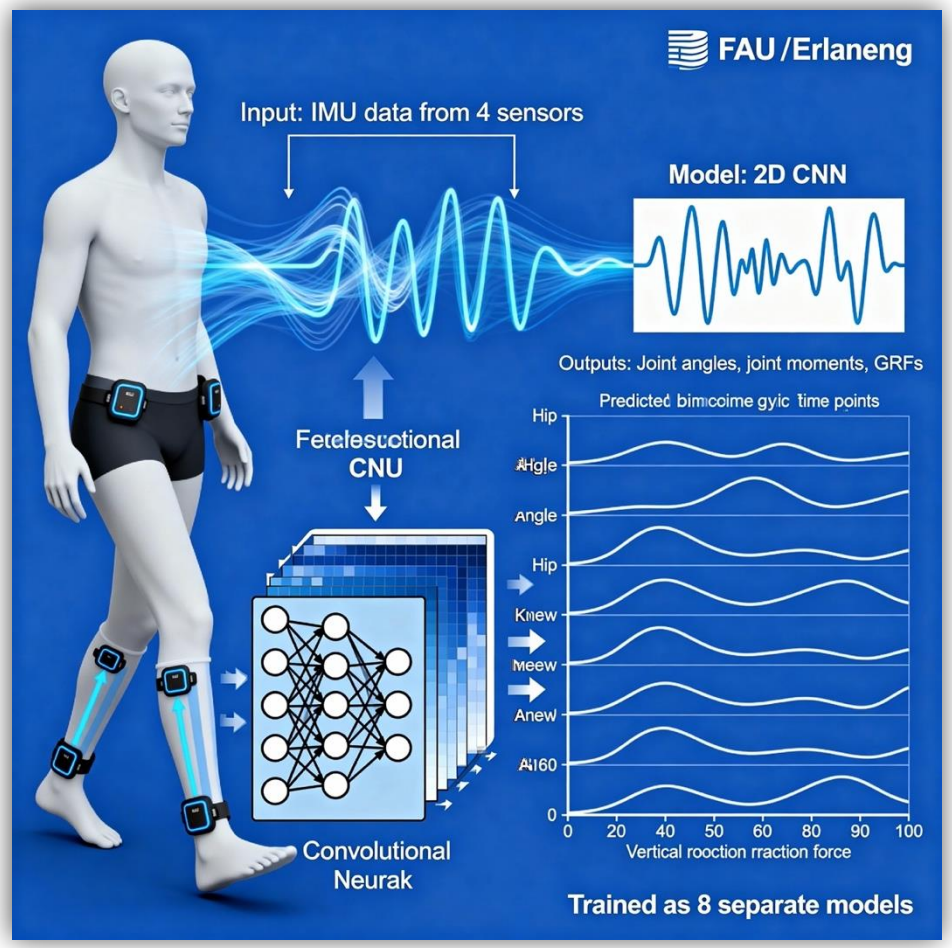
CNN Model Overview

✓ Architecture Choice

- They tested different architectures
- **2D CNN layers** performed better than 1D CNNs
  - likely due to richer temporal–sensor feature extraction
- They trained **eight separate CNNs Models**
  - One for each target variable

✓ This helps to improve stability and avoids interference between different outputs



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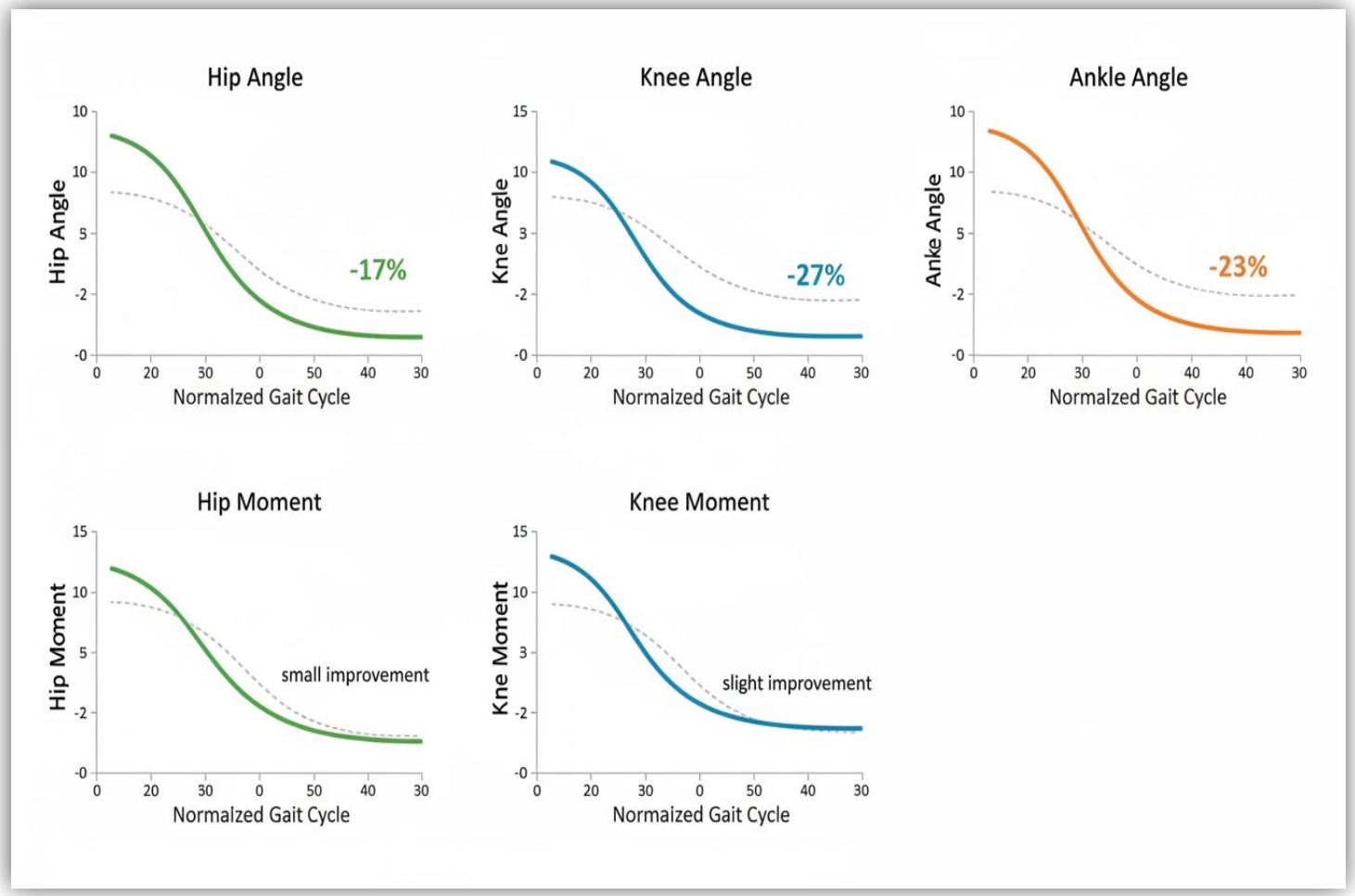
Results

- ✓ Overall Findings
  - Adding simulated data significantly improved joint angle estimation
    - Hip angle
    - Knee angle
    - Ankle angle
  - Joint Moments
    - Hip moment sometimes performed best without simulation
    - Knee moment improved slightly with simulated data



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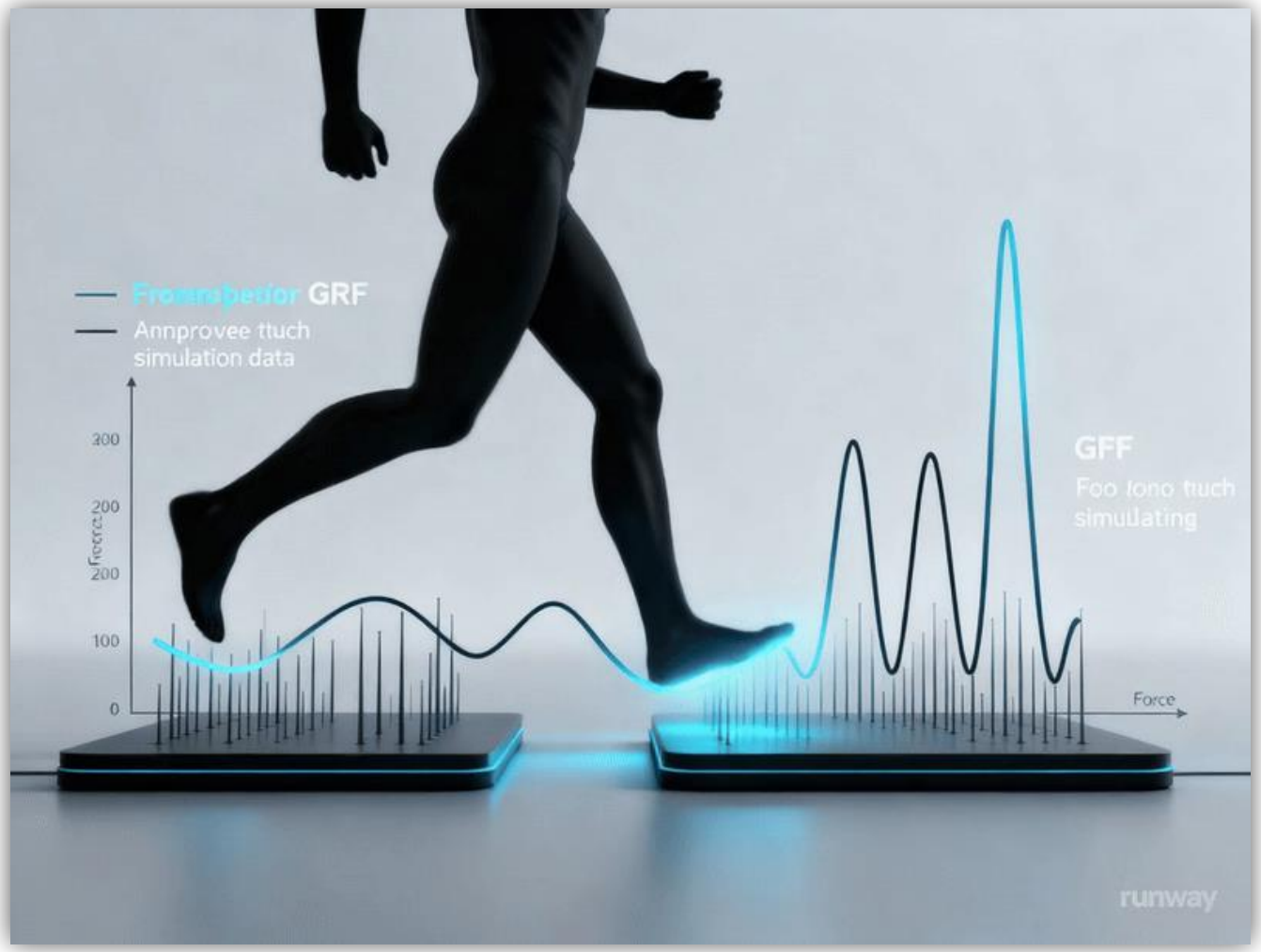
Results

- Ground Reaction Forces (GRFs)
  - Improvements for **A–P GRF** and **Vertical GRF** in some subjects
  - But performance **decreased** when too much simulated data was added
  - **Too much simulated data leads to overfitting**, especially for GRFs



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Discussion

- What This Paper Achieved

1- Hybrid Data Approach

- ✓ Significantly improves model generalization

2- Simulation Pipeline

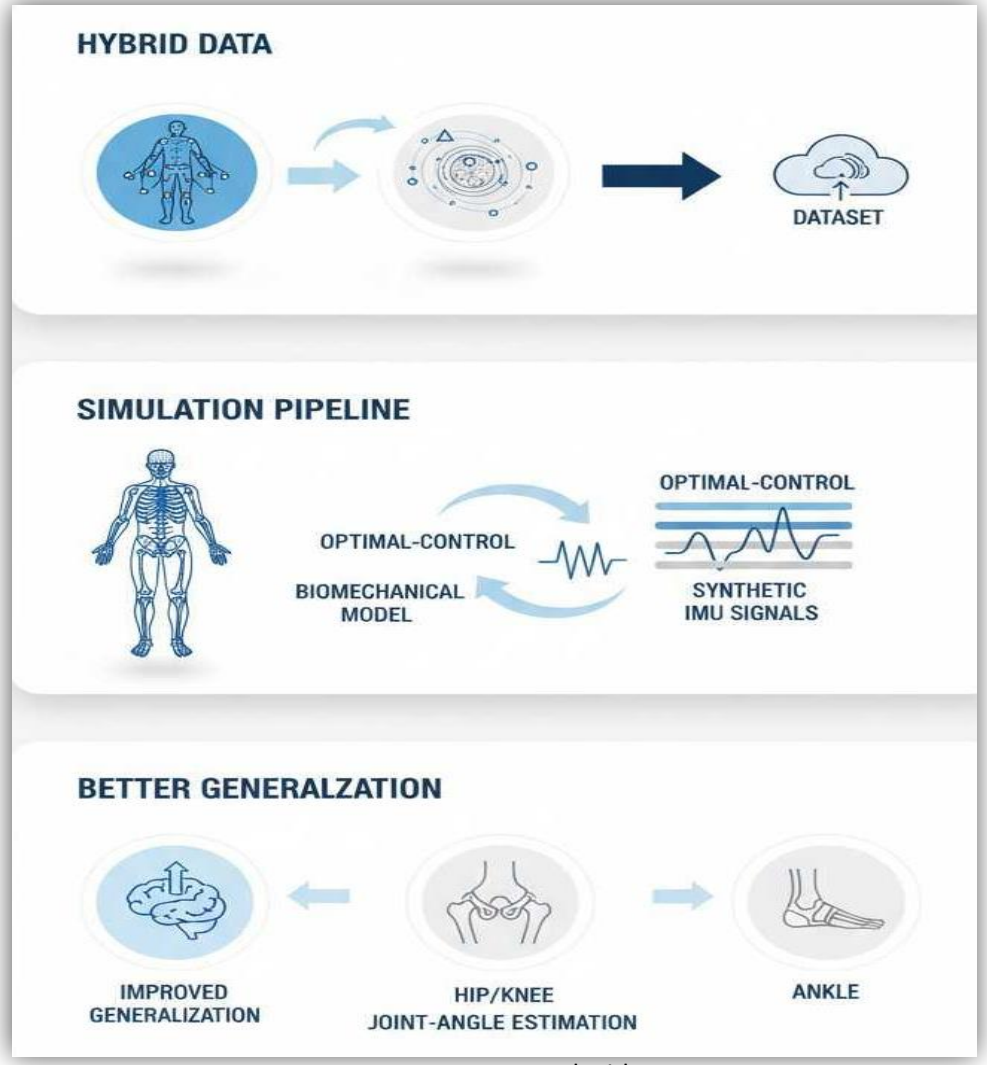
- ✓ Helps overcome the **small dataset problem** in biomechanics

3- Improved Biomechanics Estimation Without Lab Equipment

- ✓ Practical for real-world use

4- Demonstrated Effect of Simulation Scale

- ✓ Showed how much simulated data **improves** accuracy
- ✓ Identified risks of **overfitting**



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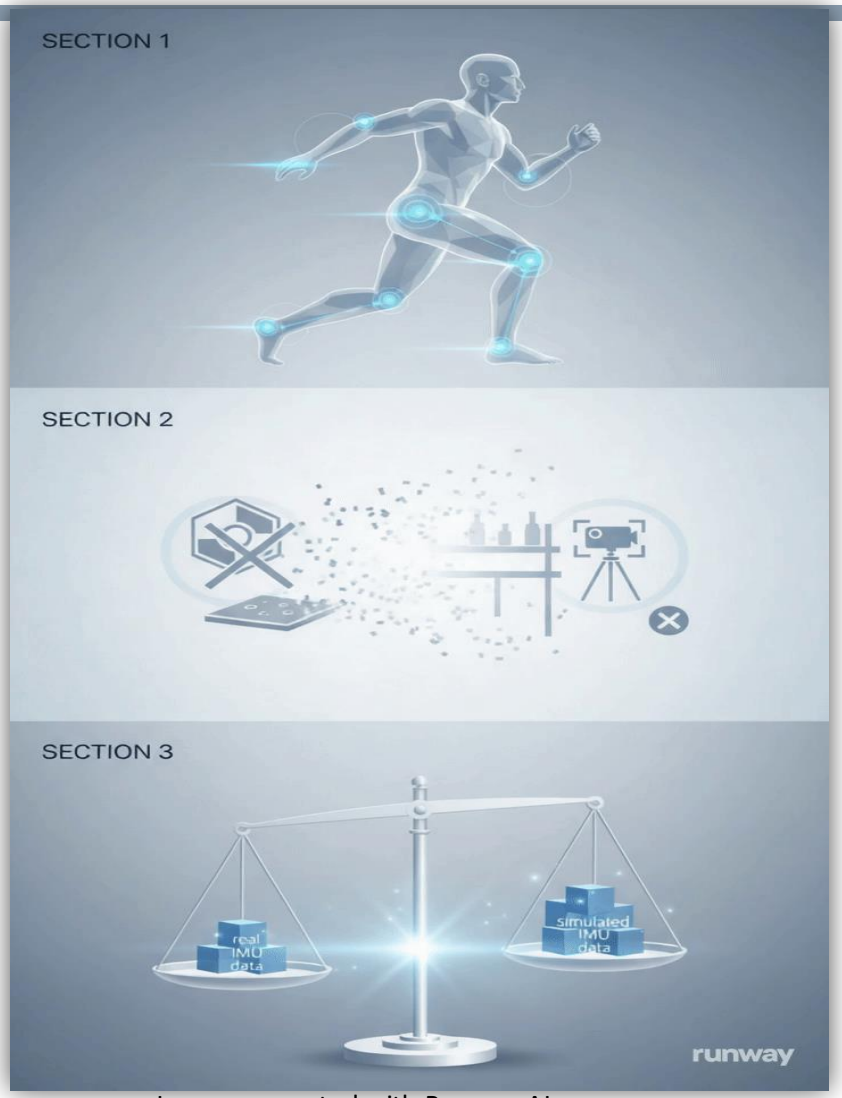
Discussion

- Limitations
  - No Noise Added
  - Simulated Data Comes from a 2D Model Only
    - This ignores out-of-plane movements, which are present in real human gait
  - Tested Only Walking and Running
    - Other movements, such as stairs or uneven terrain, are not considered



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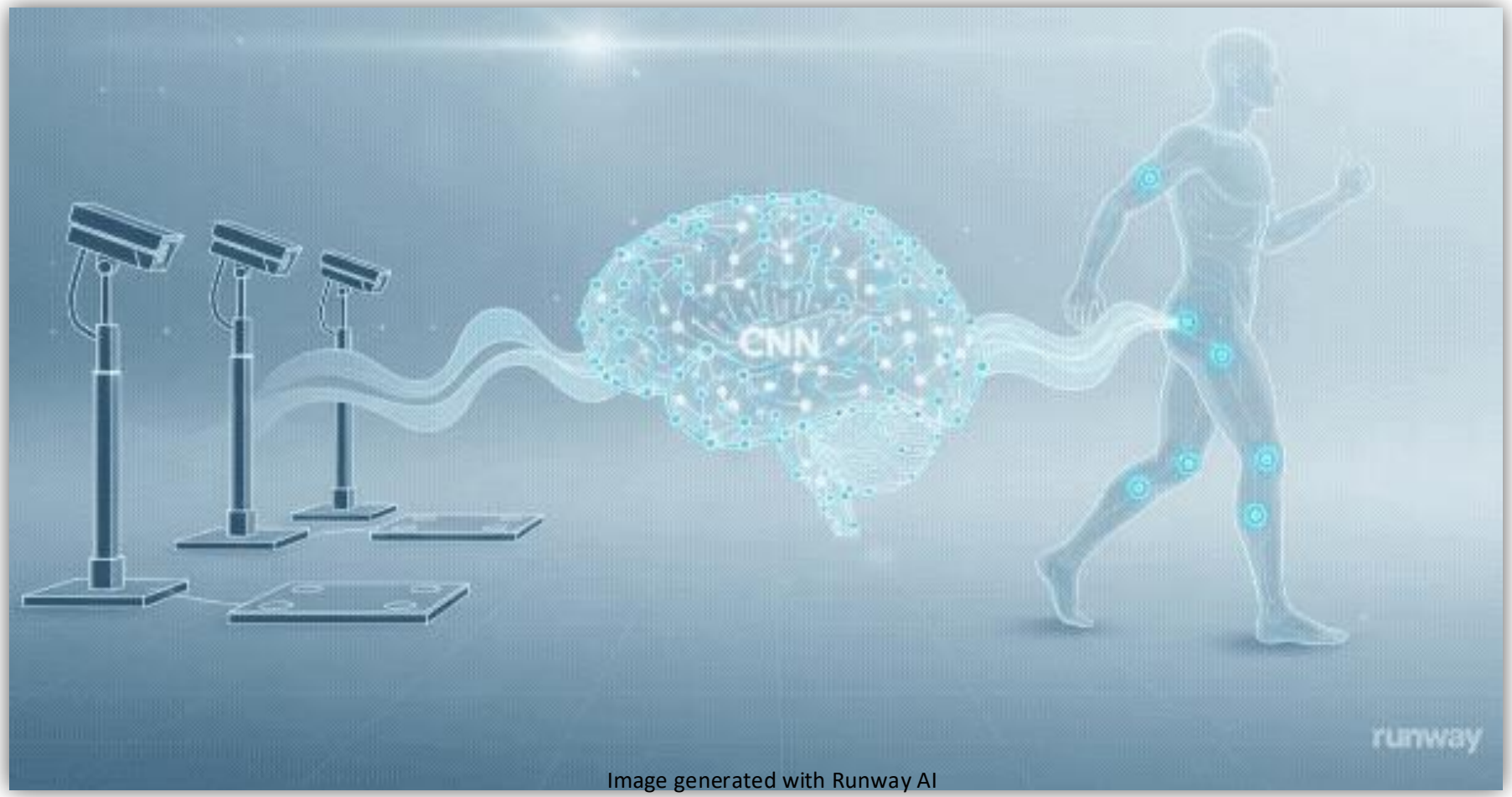
Conclusion



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This paper demonstrates that **wearable IMUs can replace expensive lab equipment** by generating realistic simulated IMU data **to overcome the small-dataset problem**, enabling a CNN to accurately predict biomechanics — and even achieve **better performance** than using real data alone.

Proposed Approach

1- Motion & Ground Truth Generation

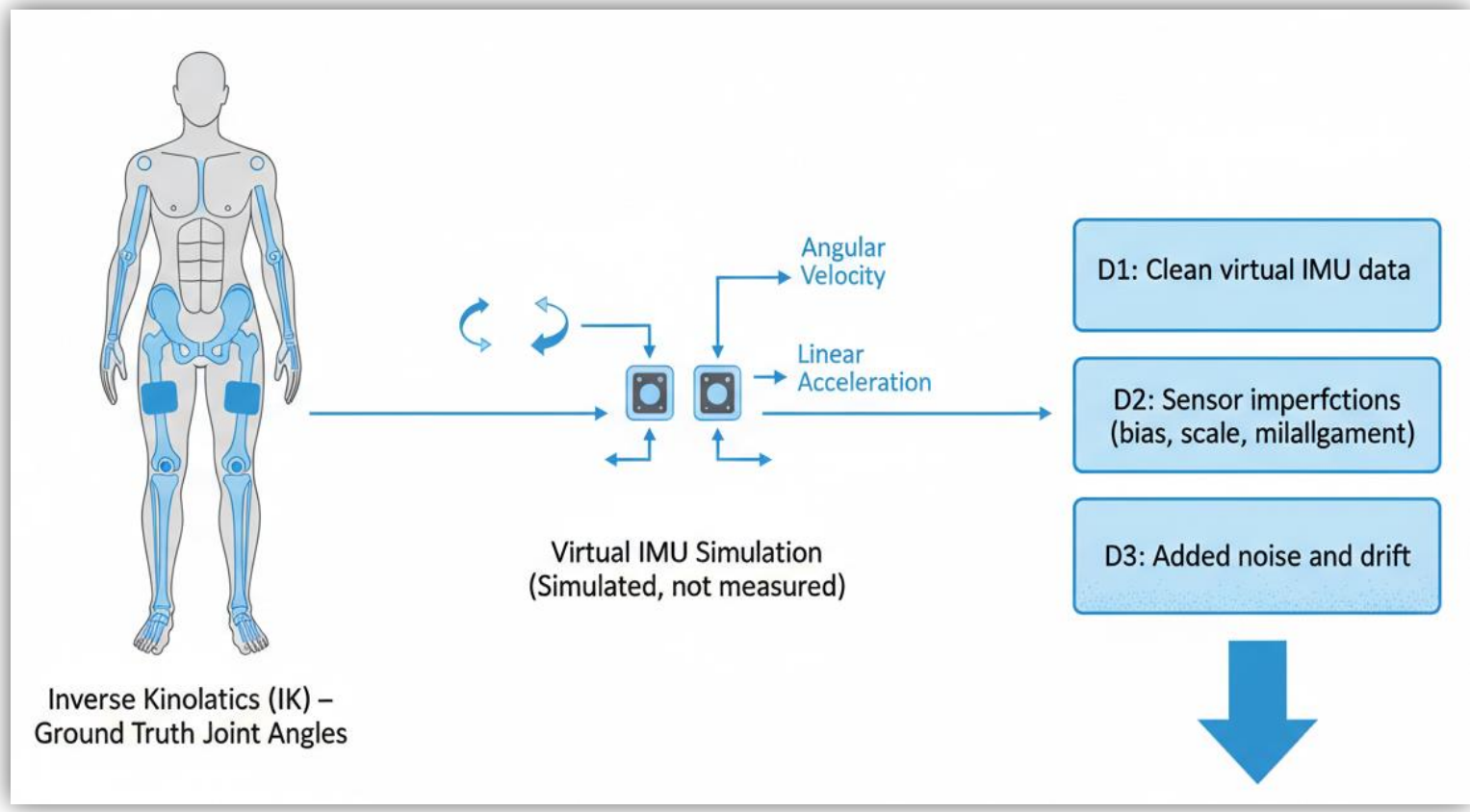
- ✓ OpenSim musculoskeletal models
- ✓ from Inverse Kinematics (IK) motion

2- Virtual IMU Simulation

- ✓ Angular velocity
- ✓ Linear acceleration
- ✓ IMU signals are simulated (not measured)

➢ Dataset Definition

- ✓ D1
- ✓ D2
- ✓ D3



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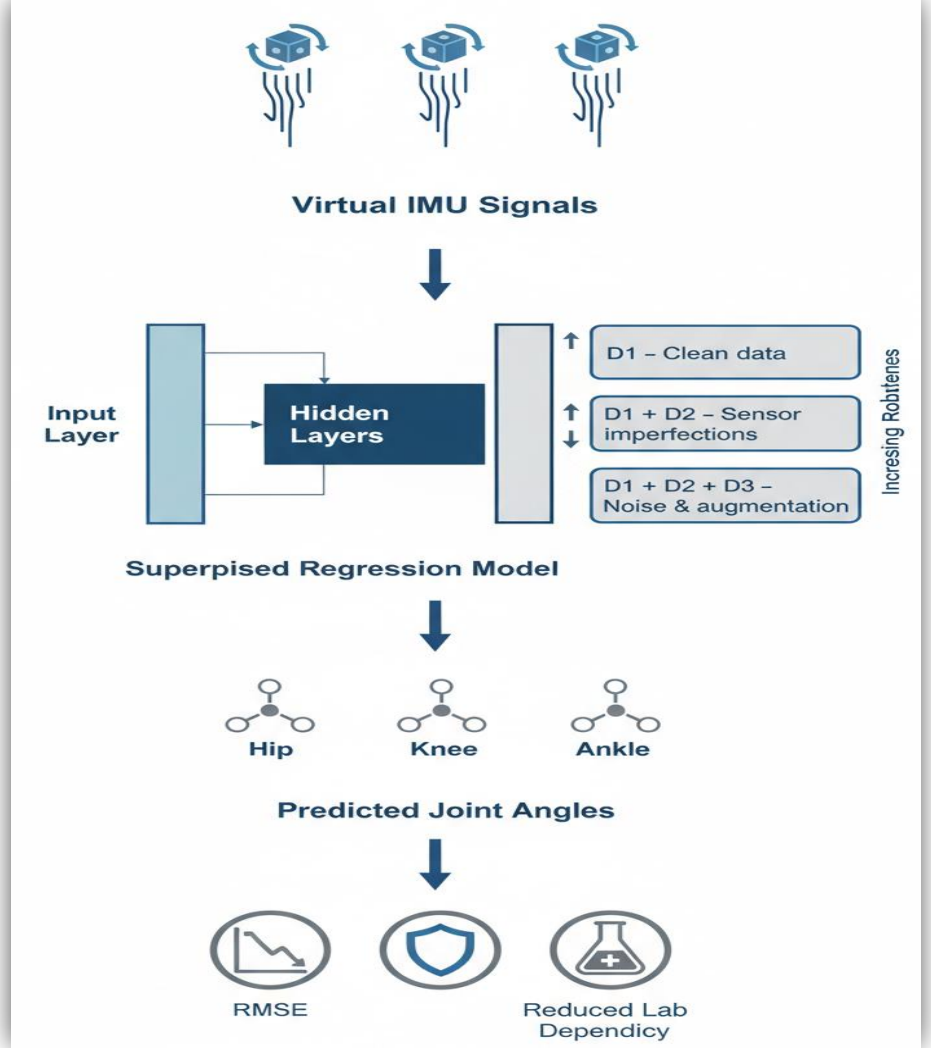
Proposed Approach

- Machine Learning Task
  - ✓ Supervised regression
  - ✓ Output: Joint angles
- Training Strategy
  - ✓ D1
  - ✓ D1 + D2
  - ✓ D1 + D2 + D3
- Evaluation
  - ✓ Compare RMSE
  - ✓ Effect of sensor imperfections
  - ✓ Impact of noise and data augmentation



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References

- Dorschky, E., Nitschke, M., Martindale, C. F., van den Bogert, A. J., Koelewijn, A. D., & Eskofier, B. M. (2020). CNN-based estimation of sagittal plane walking and running biomechanics from measured and simulated inertial sensor data. Frontiers in Bioengineering and Biotechnology, 8, 604. Available at: https://www.frontiersin.org/journals/bioengineering-and-biotechnology/articles/10.3389/fbioe.2020.00604/full?utm
- Dorschky, E., et al. (2020). Dataset for “CNN-based estimation ...”. Zenodo. Record: https://zenodo.org/records/11522050
- Dorschky, E., et al. (2020). Supplementary and full PDF: file:///mnt/data/202005_frontiers_dorschky.pdf

Thank you for your attention :)